Time-Series-Driven Predictive Maintenance with NASA C-MAPSS

Stat 248 Final Project

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Abstract

This project studies predictive maintenance as a time-series forecasting problem using the NASA C-MAPSS FD001 turbofan engine degradation benchmark. The main question is whether richer temporal summaries improve short-horizon failure-risk prediction and, in turn, lower maintenance cost. Results compare generalized linear baselines, a nonlinear time-series forest, and several maintenance policies driven by predicted risk. In the main snapshot-only versus temporal-context comparison, the strongest predictive model is the nonlinear time-series forest, while a tuned risk-threshold policy achieves the lowest maintenance cost.

1 Introduction

Predictive maintenance is a natural time-series problem because engine failure risk depends on how sensor signals evolve over time rather than on one isolated reading. This project evaluates whether explicit temporal features from C-MAPSS trajectories improve near-term failure prediction and whether those better risk estimates translate into better maintenance decisions.

2 Data

The analysis uses the NASA C-MAPSS FD001 subset. Because the downstream maintenance experiment requires complete run-to-failure trajectories, engine units from the training file are split into train, validation, and test partitions. The supervised label at each cycle is whether failure occurs within the next thirty cycles.

The repository also includes the official truncated C-MAPSS test files and their associated RUL labels. Those files are appropriate for prediction-only benchmarking, but they are not used as the primary evaluation split here because the maintenance-policy layer requires full trajectories that can be rolled forward to failure.

Table 1: Summary of the FD001 engine-unit splits used in the analysis.

split	n_rows	n_machines	mean_failure_time	mean_path_length	positive_rate_h30
train	14520	70	207.4286	207.4286	0.1494
validation	3001	15	200.0667	200.0667	0.1549
test	3110	15	207.3333	207.3333	0.1495

3 Methods

Three predictive models are compared. The baseline GLM uses only current-cycle operating settings and sensor readings from sensor_7, sensor_11, sensor_12, sensor_15, sensor_20, sensor_21. The advanced GLM adds rolling means, rolling standard deviations, slopes, and Holt summaries. The nonlinear time-series forest extends this feature set with lags, first and second differences, exponentially weighted averages, and short-run versus long-run sensor contrasts.

These are appropriate time-series models for the project goal because the task is supervised near-failure event prediction from multivariate degradation streams rather than one-step-ahead forecasting of a single stationary series. In that setting, engineered temporal summaries and nonlinear interactions are a natural way to preserve trend, persistence, and local acceleration information while keeping the workflow interpretable and reproducible.

For the nonlinear model, a small validation-based hyperparameter sweep over tree depth, leaf size, feature subsampling, and number of trees is used to improve out-of-sample performance without shifting the project toward large-scale automated search.

To connect prediction with decision-making, the project compares reactive maintenance, a fixed-age replacement rule, a tuned risk-threshold policy, tabular Q-learning, and a compact DQN extension. The strengthened RL state includes current predicted risk, age, and recent risk change, and the training reward discourages continuing operation under high predicted risk. The RL methods are included as benchmarks, but the main focus remains on the time-series forecasting layer.

4 Predictive Results

Table 2: Predictive performance on held-out FD001 engine units.

model	auc	pr_auc	brier	log_loss
nonlinear_ts_forest	0.9962	0.9797	0.0193	0.0635
advanced_glm	0.9936	0.9680	0.0227	0.0739
baseline_glm	0.9808	0.9178	0.0371	0.1234

The best predictive model is `nonlinear_ts_forest`. Because the main comparison is snapshot-only versus temporal-context models, this result supports the claim that explicit time-series information improves short-horizon failure prediction. Grouped bootstrap confidence intervals by engine unit are additionally saved in `outputs/risk_model_metrics_with_ci.csv`.

Table 3: Validation results for the nonlinear time-series forest hyperparameter sweep.

candidate	n_estimators	max_depth	min_samples_leaf	max_features	auc	pr_auc	brier	log_loss
4	300	10	6	0.5	0.9930	0.9674	0.0235	0.0781
5	300	12	6	0.5	0.9930	0.9672	0.0235	0.0781
3	240	10	8	sqrt	0.9936	0.9701	0.0235	0.0797
1	180	8	10	sqrt	0.9936	0.9697	0.0237	0.0804
2	240	8	8	sqrt	0.9935	0.9695	0.0237	0.0803

Table 4: Top ten feature importances for the nonlinear time-series forest.

feature	importance
sensor_11_mean_5	0.3179
sensor_11_ewm_04	0.2485
mean_sensor_level_5	0.0570
sensor_11_slope_30	0.0564
holt_level	0.0370
sensor_11_mean_15	0.0297
sensor_15_mean_5	0.0232
sensor_15_mean_15	0.0199
sensor_12_slope_30	0.0186
sensor_15_ewm_04	0.0147

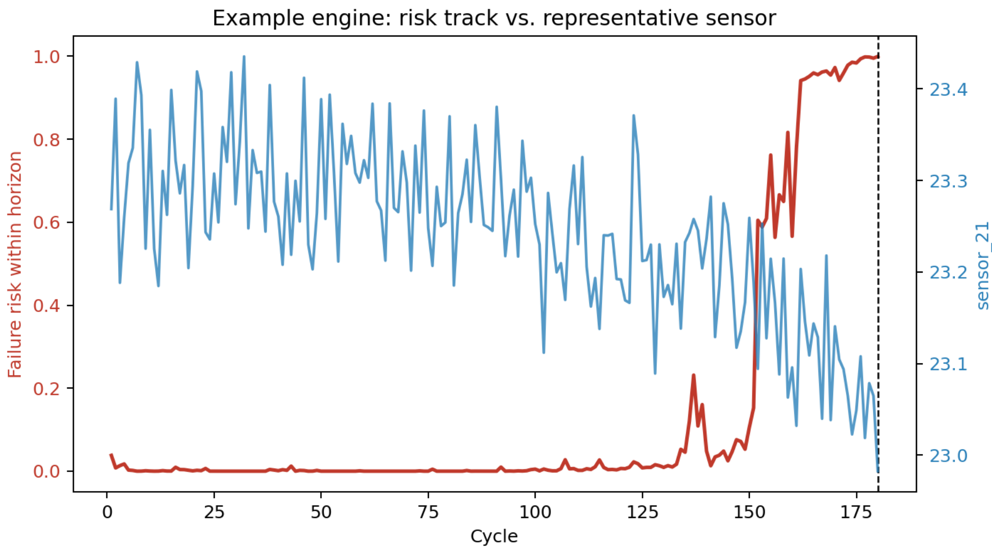

Figure 1: Predicted short-horizon failure risk for one example engine trajectory, shown alongside a representative sensor channel.

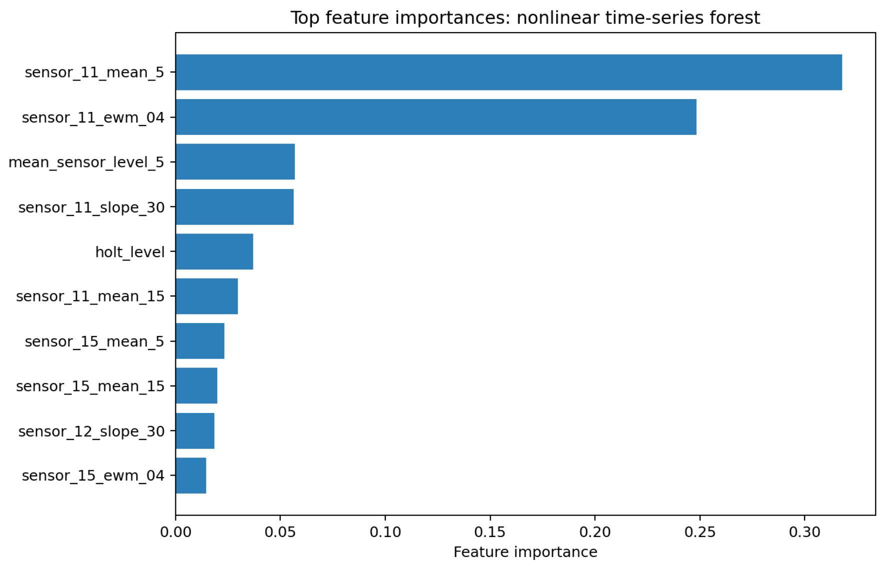

Figure 2: Top feature importances for the nonlinear time-series forest.

5 Diagnostics

To validate the time-series workflow, the project includes augmented Dickey-Fuller checks on first-differenced representative sensor channels, calibration summaries for predicted probabilities, and residual dependence diagnostics.

Table 5: ADF diagnostics on first-differenced representative sensor channels.

series	adf_stat	p_value	crit_5pct
sensor_7_first_difference	-40.8928	0.0000	-2.8617
sensor_12_first_difference	-48.5081	0.0000	-2.8617
sensor_21_first_difference	-20.7355	0.0000	-2.8617

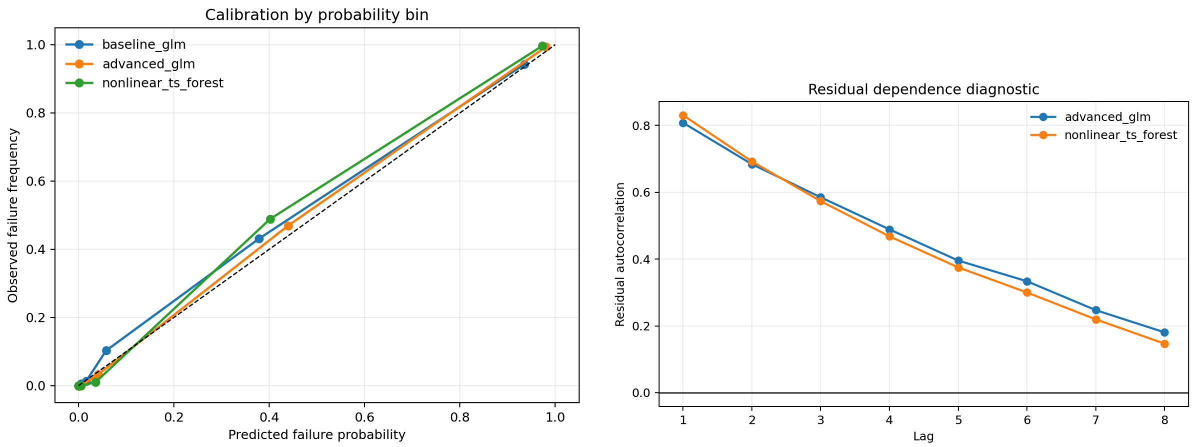

Figure 3: Calibration and residual dependence diagnostics for the strongest predictive models.

6 Maintenance Policy Comparison

Policies are evaluated in a continuing-operation environment that repeatedly samples held-out FD001 trajectories. Preventive replacement costs 18 units, while failure-driven replacement costs 95 units. The risk-threshold rule is tuned on the validation split by sweeping candidate thresholds and selecting the one with the lowest expected cost.

Table 6: Average cost and event counts by maintenance policy.

policy	avg_reward	avg_cost	std_cost	avg_replacements	avg_failures
risk_threshold_0.30	-414.0000	414.0000	15.2128	23.0000	0.0000
age_threshold_196	-1108.2143	1108.2143	152.2292	20.9286	9.5000
q_learning	-1210.9286	1210.9286	103.4377	22.3571	10.5000
reactive	-1784.6429	1784.6429	81.7108	18.7857	18.7857
dqn	-1784.6429	1784.6429	73.3987	18.7857	18.7857

The tuned risk-threshold rule uses a threshold of 0.30. It achieves the lowest average cost in this experiment, which reinforces the practical story of the project: once the risk estimate is informative, a simple decision rule can be highly effective.

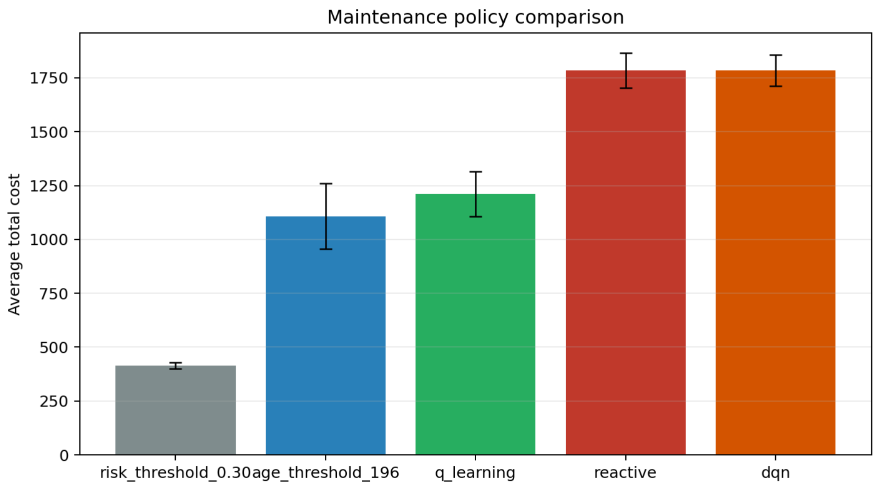

Figure 4: Average maintenance cost across competing policies.

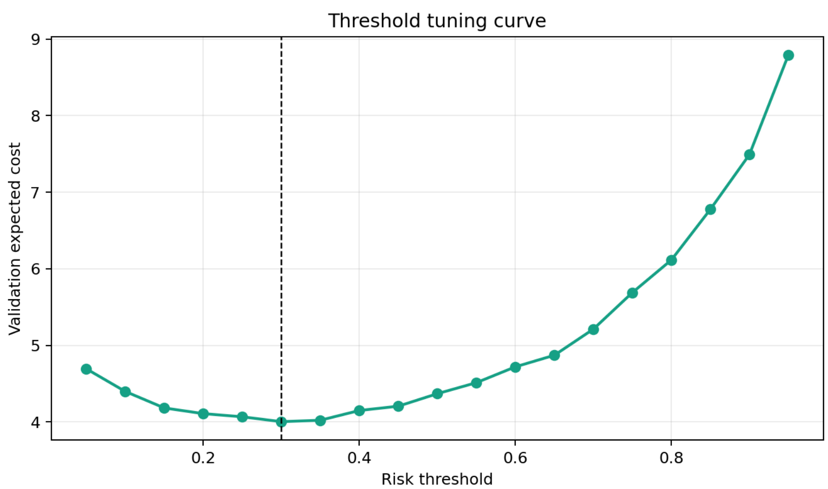

Figure 5: Validation threshold sweep for the risk-based maintenance policy.

7 Robustness Extension

To assess whether the central forecasting result depends too strongly on the simplest C-MAPSS subset, the predictive comparison was rerun on FD002, FD003, and FD004 using the same train-validation-test protocol. This extension is limited to prediction rather than full policy evaluation so that it can function as an appendix-style robustness check.

Table 7: Predictive robustness across the four standard C-MAPSS subsets.

subset	model	auc	pr_auc	brier	log_loss
FD001	baseline_glm	0.9808	0.9178	0.0371	0.1234
FD001	advanced_glm	0.9936	0.9680	0.0227	0.0739
FD001	nonlinear_ts_forest	0.9962	0.9797	0.0193	0.0635
FD002	baseline_glm	0.9380	0.7986	0.0598	0.2062
FD002	advanced_glm	0.9823	0.9111	0.0368	0.1246
FD002	nonlinear_ts_forest	0.9833	0.9181	0.0583	0.2144
FD003	baseline_glm	0.9872	0.9133	0.0282	0.0938
FD003	advanced_glm	0.9962	0.9741	0.0164	0.0520
FD003	nonlinear_ts_forest	0.9966	0.9780	0.0148	0.0501
FD004	baseline_glm	0.9161	0.7321	0.0503	0.1868
FD004	advanced_glm	0.9799	0.8566	0.0370	0.1185
FD004	nonlinear_ts_forest	0.9489	0.7213	0.0611	0.2194

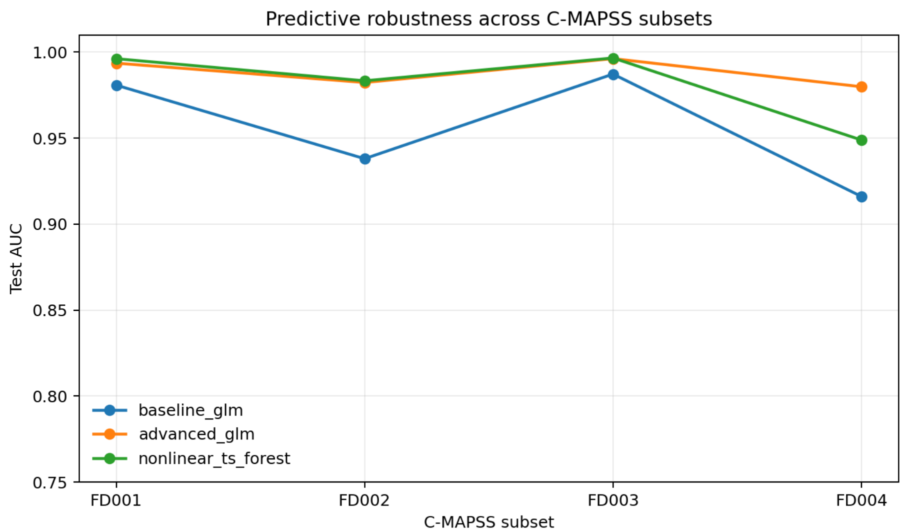

Figure 6: AUC comparison across FD001 through FD004.

8 Discussion

The strongest practical lesson in this project is that temporal context improves prediction when compared against a true snapshot-only baseline. The richer models capture information in short-run and medium-run sensor evolution that is not available from a single cycle alone.

9 Limitations and Future Work

The main limitations are that the operational comparison is still lightweight and that the robustness extension focuses on predictive metrics rather than full policy evaluation on every subset. Even after strengthening the RL state and reward, the tuned threshold controller remains easier to tune and more effective in this setup. Natural extensions include subset-specific operating regimes, richer partial observability, and recurrent or state-space formulations for maintenance control.

10 Reproducibility

The full pipeline can be reproduced by placing the NASA C-MAPSS files in `data/CMAPSSData/`, installing dependencies from `requirements.txt`, and running `python run_project.py`. The script regenerates all tables, figures, and both report formats.